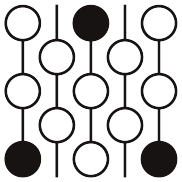
Review of Problem 3 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

The overall solution strategy is very similar to the human solution and the Otter solution, but it is notably lower quality and would require a major rewrite to be appropriate for a journal submission.

Correctness. I do not find any errors or false statements, but some important steps are not satisfactorily justified and some statements are imprecise enough that I cannot say they are correct. An expert could salvage the solution without changing the overall structure of the document.

Exposition. There are numerous places where the solution references facts in ways that are either not precise or potentially misleading. For example, the abstract mentions an "ordered two-arc factorization theorem" when no theorem has such a name.

Attribution. The solution is very light on attributions, but possibly gives a minimal set of references in order to understand the solution of the problem. However, the $n = 2$ solution is in the literature and is not referenced.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

We give an explicit identification of the reduced stated SL_n-skein algebra of the once-punctured bigon with the generalized quantum double of the two simply-connected quantum Borel algebras. The identification is specified on every horizontal matrix-coefficient arc.

> **Reviewer 1**
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> The phrase mixes and matches the phrases “horizontal arc" and "matrix-coefficient" in an unnatural way.

Its injectivity follows from an ordered two-arc factorization of the skein algebra, rather than from a comparison of defining relations alone. The prescribed diagonal

> **Reviewer 1**
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> This sentence is not appropriate for the abstract since the reader does not know what has been “prescribed" in the problem statement.

quotient then identifies the two Cartan tori and gives the simply-connected quantized enveloping algebra.

3 Coefficients and conventions

Fix $n \geq 2$. We give the proof over

$$\mathbb{K} = \mathbb{Q}(v), \qquad q = v^{2n}, \qquad \lambda = q - q^{-1}.$$

Thus all fractional powers needed in the stated skein relations are available, and λ is invertible. The same argument applies at non-root-of-unity specializations over a common ground field containing the required fractional powers. Throughout, $\mathcal{U}_q(\mathfrak{sl}_n)$ means the unrestricted, simply-connected quantum group, not a small quantum group.

We use the stated skein conventions of Lê–Sikora [3]: the product xy places x above y. The reduced algebra is the quotient by the bad corner arcs, as in [4, Section 7.1].

The surface $\mathfrak{B}^*$ is a disk with its top and bottom boundary points and one interior point removed. We fix the following labeling of the horizontal arcs. Both arcs point to the right; the first index is the left state and the second is the right state.

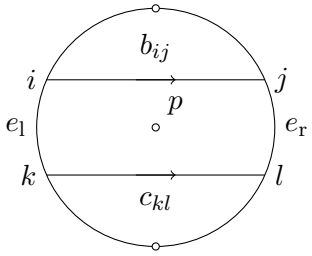


Figure 1: The labeling used in the formulas below.

At the upper corner the bad right-directed arcs are b_{ij} with $i < j$. At the lower corner they are c_{ij} with $i > j$. Consequently,

$$b_{ij} = 0 \quad (i < j), \qquad c_{ij} = 0 \quad (i > j) \qquad \text{in } \overline{\mathscr{S}}_n(\mathfrak{B}^*). \tag{1}$$

The reversal of the inequality at the lower corner is due to the 180° rotation identifying that corner with the standard upper-corner picture. If the two letters in a different drawing are interchanged, the formulas below are to be read with that interchange.

For the quantum group, put

$$\omega_{ai} = \delta_{a,i} - \delta_{a,i+1}, \qquad K_i = \widehat{K}_i \widehat{K}_{i+1}^{-1}.$$

The simply-connected presentation uses

$$\widehat{K}_a^{\pm 1} \quad (1 \le a \le n), \qquad E_i, F_i \quad (1 \le i < n),$$

with commuting toral generators and

$$\prod_{a=1}^{n} \widehat{K}_a = 1, \tag{2}$$

$$\widehat{K}_a E_i \widehat{K}_a^{-1} = q^{\omega_{ai}} E_i, \qquad \widehat{K}_a F_i \widehat{K}_a^{-1} = q^{-\omega_{ai}} F_i, \tag{3}$$

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{\lambda}. \tag{4}$$

For $X = E$ and $X = F$ the remaining relations are

$$X_i X_j = X_j X_i \qquad (|i-j| > 1), \tag{5}$$

$$X_i^2 X_j - (q + q^{-1}) X_i X_j X_i + X_j X_i^2 = 0 \qquad (|i-j| = 1). \tag{6}$$

Our coproduct convention is

$$\begin{aligned} \Delta(\widehat{K}_a) &= \widehat{K}_a \otimes \widehat{K}_a, \\ \Delta(E_i) &= E_i \otimes K_i + 1 \otimes E_i, \\ \Delta(F_i) &= F_i \otimes 1 + K_i^{-1} \otimes F_i. \end{aligned} \tag{7}$$

These conventions specify the extended quantum group in the question; see [2]. The corresponding Borel presentations use the toral relations and the respective E- or F-relations. They also follow from the quantum-matrix realizations recalled in Lemma 5.1 below.

4 The skein-theoretic matrix calculation

Let $V = \mathbb{K}^n$, let e_{ab} denote its matrix units, and set

$$R = q^{-1/n} \left(\sum_{a,b=1}^{n} q^{\delta_{ab}} e_{aa} \otimes e_{bb} + \lambda \sum_{a>b} e_{ab} \otimes e_{ba} \right). \tag{8}$$

The braiding matrix is PR, where P interchanges the tensor factors. This is the normalization used in [3, Equation (9)].

> **Reviewer 1**
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> This definition should be more precise in constructing the algebra by generators and relations. The t_{ij} should be identified as generators of the algebra and the matrix equations should be identified as the defining relations of the algebra.

For a matrix $T = (t_{ij})$, write $T_1 = T \otimes I_n$ and $T_2 = I_n \otimes T$. The standard quantum coordinate algebra $\mathcal{O}_q(\mathrm{SL}_n)$ has presentation

$$RT_1T_2 = T_2T_1R, \qquad \det_q(T) = 1, \tag{9}$$

where

$$\det_q(T) = \sum_{\sigma \in S_n} (-q)^{\ell(\sigma)} t_{1,\sigma(1)} \cdots t_{n,\sigma(n)}.$$

Define its triangular quotients

$$\mathcal{T}^- = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i < j), \qquad \mathcal{T}^+ = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i > j). \tag{10}$$

Lemma 4.1 (Two-arc matrix calculation)**.** *Let $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$, and let $B = (b_{ij})$ and $C = (c_{ij})$ be as in Figure 1.*

1. *B and C satisfy* (9)*. Moreover, B is lower triangular and C is upper triangular.*
2. *The mixed relation is*

$$RB_1C_2 = C_2B_1R. \tag{11}$$

   Equivalently, for all i, j, k, l,

$$q^{\delta_{ik}} b_{ij}c_{kl} - q^{\delta_{jl}} c_{kl}b_{ij} = \lambda\big(\mathbf{1}_{j<l} c_{kj}b_{il} - \mathbf{1}_{k<i} b_{kj}c_{il}\big). \tag{12}$$

3. *The homomorphisms from $\mathcal{T}^-$ and $\mathcal{T}^+$ determined by $T \mapsto B$ and $T \mapsto C$ are injective, and multiplication induces a vector-space isomorphism*

$$\mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ \xrightarrow{\sim} A, \qquad f \otimes g \longmapsto f(B)g(C). \tag{13}$$

   This is a vector-space assertion, not an assertion that the ordinary tensor-product multiplication is preserved.

Proof. A thin neighborhood of either horizontal arc is an ideal bigon. The bigon theorem [3, Theorem 6.3] identifies its stated skein algebra with $\mathcal{O}_q(\mathrm{SL}_n)$, with the right-directed stated arcs as the matrix generators. This proves the quantum-matrix and determinant relations. Equation (1) gives the asserted triangularity.

Here are the boundary-exchange details for the mixed relation. They hold already in the unreduced algebra. Consider the two disjoint arcs with c higher than b at the left boundary and b higher than c at the right boundary. Denote the resulting mixed-height diagram, with external state pairs (i, k) and (j, l), by $W_{ik;jl}$. Applying the boundary height-exchange relation at the left or at the right gives, respectively,

$$W_{ik;jl} = \sum_{a,b} R_{ik,ab}\, b_{aj}c_{bl} = \sum_{a,b} c_{kb}b_{ia}\, R_{ab,jl}. \tag{14}$$

Here the flip in the braiding PR has been removed by recording the indices in the order of the two arcs. Thus the coefficients appearing in the two exchanges are

$$R_{ik,ab} = q^{-1/n}\left(q^{\delta_{ik}}\delta_{ia}\delta_{kb} + \lambda \mathbf{1}_{k<i}\delta_{ka}\delta_{ib}\right),$$
$$R_{ab,jl} = q^{-1/n}\left(q^{\delta_{jl}}\delta_{aj}\delta_{bl} + \lambda \mathbf{1}_{j<l}\delta_{al}\delta_{bj}\right).$$

These are precisely the boundary braiding coefficients (8); the boundary relations and their braiding interpretation are those of [3, Section 4 and Theorem 6.4]. Substitution into (14), followed by cancellation of $q^{-1/n}$, proves (12).

It remains to prove the vector-space assertion, including the absence of additional relations after reduction. Write $S = \mathscr{S}_n(\mathfrak{B}^*)$ for the unreduced algebra, and temporarily use b, c for the unreduced arcs. They form the standard saturated

> **Reviewer 1**
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> The phrase “saturated system of arcs" is specialized, and its definition or a reference to [3] must be mentioned at this point in the exposition.

two-arc system for $\mathfrak{B}^*$: together with the boundary edges they give a groupoid spine of the punctured bigon. The ordered-arc factorization theorem

> **Reviewer 1**
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> No theorem goes by this name. A description of the theorem statement or a precise reference to a numbered theorem in the literature is required.

, with the b-neighborhood above the c-neighborhood at both boundary edges, gives

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> There is more than one notion of “above" when talking about the stated skein algebra of the punctured bigon. It could refer to the height in the product structure or it could refer to the arc being above or below the puncture when drawn in the bigon. Figure 1 should be referenced here.

$$m : \mathcal{O}_q(\mathrm{SL}_n) \otimes_{\mathbb{K}} \mathcal{O}_q(\mathrm{SL}_n) \xrightarrow{\sim} S, \qquad f \otimes g \longmapsto f(B)g(C). \tag{15}$$

The relevant result permits arbitrary orders at the boundary edges; see [3, Corollary 8.14].

Let J_b be the ideal of the first copy of $\mathcal{O}_q(\mathrm{SL}_n)$ generated by t_{ij} with $i < j$, and let J_c be the ideal of the second copy generated by t_{ij} with $i > j$. Let I_{top} and I_{bottom} be the respective bad-corner ideals in S.

The bad-arc normality theorem [4, Theorem 7.2, especially Equation (128)] says that each corner ideal is both the left and the right ideal generated by either one of its two oriented families of bad arcs. In our notation this gives

$$I_{\mathrm{top}} = \mathrm{span}_{\mathbb{K}}\{b_{ij} : i < j\}\, S, \qquad I_{\mathrm{bottom}} = S\, \mathrm{span}_{\mathbb{K}}\{c_{ij} : i > j\}.$$

In particular, reversed orientations introduce no further ideal.

Using (15), the first equality implies

$$I_{\mathrm{top}} \subseteq m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The reverse inclusion holds because I_{top} is a two-sided ideal containing the generators of J_b. Hence

$$I_{\mathrm{top}} = m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The same argument, using the displayed right-ideal description at the lower corner, gives

$$I_{\text{bottom}} = m(\mathcal{O}_q(\mathrm{SL}_n) \otimes J_c).$$

Therefore

$$I^{\text{bad}} = m\big(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n) + \mathcal{O}_q(\mathrm{SL}_n) \otimes J_c\big).$$

Taking the quotient of the vector-space isomorphism (15) proves (13). The injectivity of the two factor maps follows as well. □

We record two elementary consequences that will also make the eventual isomorphism explicit on all the arcs. The diagonal entries of either triangular quantum matrix commute, and

$$\prod_{a=1}^{n} b_{aa} = \prod_{a=1}^{n} c_{aa} = 1. \tag{16}$$

Thus every diagonal entry is invertible. Expanding the quantum-matrix relation on the indicated 2×2 submatrices gives, for $i < j - 1$,

$$b_{ji} = \lambda^{-1} b_{j-1,j-1}^{-1} [b_{j-1,i}, b_{j,j-1}], \tag{17}$$

$$c_{ij} = \lambda^{-1} c_{j-1,j-1}^{-1} [c_{i,j-1}, c_{j-1,j}]. \tag{18}$$

Indeed, these are the relations $ad - da = \lambda bc$ for rows $j-1, j$ and columns $i, j-1$, and for rows $i, j-1$ and columns $j-1, j$, respectively. In each case the diagonal factor on the right commutes with the opposite off-diagonal factor. Consequently, each triangular algebra is generated by its diagonal and immediately adjacent off-diagonal entries.

5 The quantum Borel algebras and their double

5.1 Triangular matrix realizations

Write

$$H_- = \mathcal{U}_q(\mathfrak{b}_-), \qquad H_+ = \mathcal{U}_q(\mathfrak{b}_+).$$

Lemma 5.1 (Borel matrix realizations)**.** *There are algebra isomorphisms*

$$\eta_- : H_- \xrightarrow{\sim} \mathcal{T}^-, \qquad \eta_+ : H_+ \xrightarrow{\sim} \mathcal{T}^+$$

given by

$$\eta_-(\widehat{K}_a) = b_{aa}^{-1}, \qquad \eta_-(F_i) = \lambda^{-1} b_{i+1,i+1}^{-1} b_{i+1,i}, \tag{19}$$

$$\eta_+(\widehat{K}_a) = c_{aa}, \qquad \eta_+(E_i) = -\lambda^{-1} c_{i,i+1} c_{i+1,i+1}^{-1}. \tag{20}$$

Proof. For $\mathfrak{gl}_n$, with independent toral generators $G_1, \dots, G_n$, these are exactly the algebra isomorphisms ζ_- and ζ_+ in [1, Section 2.4]. Their inverses on adjacent entries are

$$t_{i+1,i} \longmapsto \lambda G_{i+1}^{-1} F_i, \qquad t_{i,i+1} \longmapsto -\lambda E_i G_{i+1},$$

and the diagonal entries map to G_a^{-1} and G_a, respectively.

Impose the central relation $\prod_a G_a = 1$. On the matrix sides its image is the relation that the product of the diagonal entries is 1. For a triangular matrix this is exactly the quantum determinant-one relation. Taking these quotients gives (19)–(20). The toral lattice is $\mathbb{Z}^n/\mathbb{Z}(1,\dots,1)$, namely the weight lattice of type A_{n-1}, so the resulting Borel algebras are the simply-connected ones.

Only the algebra assertion of [1, Section 2.4] is used: the coproduct convention there is opposite to (7). □

5.2 A presentation of the generalized double

Let

$$D = H_- \bowtie H_+$$

be the generalized quantum double for the pairing in the question. We denote its two toral copies by $\widehat{K}_a^-$ and $\widehat{K}_a^+$, and set

$$K_i^- = \widehat{K}_i^-(\widehat{K}_{i+1}^-)^{-1}, \qquad K_i^+ = \widehat{K}_i^+(\widehat{K}_{i+1}^+)^{-1}.$$

The superscript $-$ labels the negative Borel copy; it does not mean an inverse.

For clarity, we specify the double convention. The skew pairing $\tau : H_+ \otimes H_- \to \mathbb{K}$ satisfies

$$\begin{aligned}\tau(xx', y) &= \tau(x, y_{(1)})\tau(x', y_{(2)}),\\ \tau(x, yy') &= \tau(x_{(1)}, y')\tau(x_{(2)}, y),\end{aligned}$$

and has generator values

$$\begin{aligned}\tau(\widehat{K}_a, \widehat{K}_b) &= q^{1/n-\delta_{ab}},\\ \tau(E_i, F_j) &= -\delta_{ij}\lambda^{-1},\\ \tau(\widehat{K}_a, F_j) &= \tau(E_i, \widehat{K}_b) = 0.\end{aligned} \tag{21}$$

The underlying vector space of D is $H_- \otimes H_+$, with multiplication identifying each factor with a subalgebra. Its exchange convention is

$$\sum \tau(x_{(1)}, y_{(1)})x_{(2)}y_{(2)} = \sum y_{(1)}x_{(1)}\tau(x_{(2)}, y_{(2)}), \qquad x \in H_+, \quad y \in H_-. \tag{22}$$

Writing this convention explicitly avoids ambiguity with opposite-double conventions.

Lemma 5.2 (Presentation of the double)**.** *The algebra D is presented by the two Borel presentations and the following mixed relations:*

> **Reviewer 1**
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> *The lemma should be more precise about what the generators and relations for D are.*

$$[\widehat{K}_a^+, \widehat{K}_b^-] = 0, \tag{23}$$

$$\widehat{K}_a^+ F_i(\widehat{K}_a^+)^{-1} = q^{-\omega_{ai}}F_i, \qquad \widehat{K}_a^- E_i(\widehat{K}_a^-)^{-1} = q^{\omega_{ai}}E_i, \tag{24}$$

$$[E_i, F_j] = \delta_{ij}\frac{K_i^+ - (K_i^-)^{-1}}{\lambda}. \tag{25}$$

Proof. First derive the relations in D. For the toral terms,

$$\tau(\widehat{K}_a, K_i) = \frac{q^{1/n-\delta_{ai}}}{q^{1/n-\delta_{a,i+1}}} = q^{-\omega_{ai}}.$$

Substitution into (22) proves (23) and (24).

For $x = E_i$ and $y = F_j$, the left and right sides of (22), using (7), are

$$E_iF_j - \delta_{ij}\lambda^{-1}K_i^+, \qquad F_jE_i - \delta_{ij}\lambda^{-1}(K_j^-)^{-1},$$

respectively. This proves (25).

To see that these are a complete set of mixed relations, let D_0 be the algebra given by the two Borel presentations and (23)–(25). The derived relations give a surjection $D_0 \to D$.

They also allow every word in D_0 to be written as a sum of words with the negative Borel generators first and the positive Borel generators second. Indeed, the toral exchanges merely introduce scalars; the exchange of E_i with F_j has a reordered term and terms with fewer E's and F's. Induction first on the number of such generators and then on the number of mixed inversions proves the assertion. Hence multiplication gives a surjection

$$H_- \otimes H_+ \longrightarrow D_0.$$

Its composition with $D_0 \to D$ is the defining vector-space isomorphism $H_- \otimes H_+ \to D$. It follows that both maps are isomorphisms, proving the presentation. □

6 The explicit isomorphism

Define elements of the two Borel factors, for $1 \le i < j \le n$, by

$$\mathsf{F}_{i,i+1} = F_i, \qquad \mathsf{F}_{ij} = \mathsf{F}_{i,j-1}F_{j-1} - q^{-1}F_{j-1}\mathsf{F}_{i,j-1} \qquad (j > i+1), \tag{26}$$

$$\mathsf{E}_{i,i+1} = E_i, \qquad \mathsf{E}_{ij} = E_{j-1}\mathsf{E}_{i,j-1} - q\,\mathsf{E}_{i,j-1}E_{j-1} \qquad (j > i+1). \tag{27}$$

These recursions fix all root-vector normalizations needed below.

Theorem 6.1. *There is an algebra isomorphism*

$$\Phi : \overline{\mathscr{S}}_n(\mathfrak{B}^*) \xrightarrow{\sim} \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$$

given on the horizontal arcs by

$$\Phi(b_{aa}) = (\widehat{K}_a^-)^{-1}, \qquad \Phi(c_{aa}) = \widehat{K}_a^+, \tag{28}$$

$$\Phi(b_{ji}) = \lambda(\widehat{K}_j^-)^{-1}\mathsf{F}_{ij}, \qquad \Phi(c_{ij}) = -\lambda\mathsf{E}_{ij}\widehat{K}_j^+ \qquad (i < j). \tag{29}$$

The other off-diagonal arcs vanish by (1).

Equivalently, the inverse isomorphism $\Psi = \Phi^{-1}$ is determined by

$$\Psi(\widehat{K}_a^-) = b_{aa}^{-1}, \qquad \Psi(\widehat{K}_a^+) = c_{aa}, \tag{30}$$

$$\Psi(F_i) = \lambda^{-1}b_{i+1,i+1}^{-1}b_{i+1,i}, \qquad \Psi(E_i) = -\lambda^{-1}c_{i,i+1}c_{i+1,i+1}^{-1}. \tag{31}$$

Proof. We first construct Ψ and prove that it is an isomorphism.

For brevity put, in $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$,

$$d_a = b_{aa}, \qquad t_a = c_{aa}, \qquad x_i = b_{i+1,i}, \qquad y_i = c_{i,i+1}.$$

The diagonal elements are invertible by (16). Lemma 5.1 gives algebra homomorphisms from the two Borel factors to A with the values (30)–(31). Thus it remains to check the mixed relations in Lemma 5.2.

The internal quantum-matrix relations and (12) give

$$[d_a, d_b] = [t_a, t_b] = [d_a, t_b] = 0 \tag{32}$$

and

$$\begin{aligned} d_a x_i &= q^{\omega_{ai}} x_i d_a, \quad & t_a x_i &= q^{-\omega_{ai}} x_i t_a, \\ d_a y_i &= q^{-\omega_{ai}} y_i d_a, \quad & t_a y_i &= q^{\omega_{ai}} y_i t_a. \end{aligned} \tag{33}$$

For example, setting the first matrix entry in (12) equal to b_{aa} makes both correction terms vanish by lower triangularity, and yields

$$d_a c_{kl} = q^{\delta_{al} - \delta_{ak}} c_{kl} d_a.$$

Setting the second entry equal to c_{aa} gives

$$t_a b_{kl} = q^{\delta_{ak} - \delta_{al}} b_{kl} t_a.$$

These are the mixed assertions in (33); the internal ones are obtained in the same way from the single-matrix relation.

Write

$$\mathcal{F}_i = \lambda^{-1} d_{i+1}^{-1} x_i, \qquad \mathcal{E}_i = -\lambda^{-1} y_i t_{i+1}^{-1}.$$

Equations (32) and (33) immediately give

$$t_a \mathcal{F}_i t_a^{-1} = q^{-\omega_{ai}} \mathcal{F}_i, \qquad d_a^{-1} \mathcal{E}_i d_a = q^{\omega_{ai}} \mathcal{E}_i.$$

Thus the mixed toral relations are satisfied.

For the remaining mixed relation, substitute

$$(i, k; j, l) = (j+1, i; j, i+1)$$

into (12). Triangularity makes both correction terms vanish unless $i = j$, so that

$$q^{\delta_{i,j+1}} x_j y_i - q^{\delta_{j,i+1}} y_i x_j = \lambda \delta_{ij} (d_{i+1} t_i - d_i t_{i+1}). \tag{34}$$

Using (33) to move the diagonal factors gives

$$\begin{aligned} [\mathcal{E}_i, \mathcal{F}_j] &= \lambda^{-2} d_{j+1}^{-1} \left(x_j y_i - q^{\delta_{j,i+1} - \delta_{i,j+1}} y_i x_j \right) t_{i+1}^{-1} \\ &= \delta_{ij} \lambda^{-1} d_{i+1}^{-1} (d_{i+1} t_i - d_i t_{i+1}) t_{i+1}^{-1} \\ &= \delta_{ij} \frac{t_i t_{i+1}^{-1} - d_i d_{i+1}^{-1}}{\lambda}. \end{aligned} \tag{35}$$

But under the proposed map,

$$K_i^+ \longmapsto t_i t_{i+1}^{-1}, \qquad (K_i^-)^{-1} \longmapsto d_i d_{i+1}^{-1}.$$

Therefore (35) is precisely (25). Lemma 5.2 now supplies an algebra homomorphism

$$\Psi : D \longrightarrow A$$

with the required values.

We prove its bijectivity without making an unsupported presentation claim about A. The following diagram commutes:

$$\begin{array}{ccc} H_- \otimes_{\mathbb{K}} H_+ & \xrightarrow{m_D} & D \\ {\scriptstyle \eta_- \otimes \eta_+}\big\downarrow \cong & & \big\downarrow {\scriptstyle \Psi} \\ \mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ & \xrightarrow{m_A} & A. \end{array}$$

The top arrow is the defining vector-space isomorphism for the generalized double. The left arrow is an isomorphism by Lemma 5.1, and the bottom arrow is an isomorphism by Lemma 4.1. Consequently Ψ is a vector-space isomorphism and hence an algebra isomorphism.

It remains to verify the formulas for its inverse on the non-adjacent entries. The inverse formulas on the diagonal and adjacent entries follow directly from (30)–(31). For completeness, the induction using (17) and (18) is as follows.

Put

$$m_a = (\widehat{K}_a^-)^{-1}, \qquad p_a = \widehat{K}_a^+.$$

For $i < j-1$, assume the asserted formulas at distance $j-i-1$. The toral weight relations give

$$\begin{aligned} &\lambda^{-1} m_{j-1}^{-1}[\,\lambda m_{j-1}\mathsf{F}_{i,j-1}, \lambda m_j F_{j-1}\,] \\ &\qquad = \lambda m_j\big(\mathsf{F}_{i,j-1}F_{j-1} - q^{-1}F_{j-1}\mathsf{F}_{i,j-1}\big) = \lambda m_j \mathsf{F}_{ij}. \end{aligned}$$

Here m_j commutes with $\mathsf{F}_{i,j-1}$ and $F_{j-1}m_{j-1} = q^{-1}m_{j-1}F_{j-1}$. This proves the lower-triangular formula.

Similarly,

$$\begin{aligned} &\lambda^{-1} p_{j-1}^{-1}[\,-\lambda \mathsf{E}_{i,j-1}p_{j-1}, -\lambda E_{j-1}p_j\,] \\ &\qquad = -\lambda\big(E_{j-1}\mathsf{E}_{i,j-1} - q\mathsf{E}_{i,j-1}E_{j-1}\big)p_j = -\lambda \mathsf{E}_{ij}p_j. \end{aligned}$$

Indeed, p_j commutes with $\mathsf{E}_{i,j-1}$, $p_{j-1}E_{j-1} = qE_{j-1}p_{j-1}$, and the total weight of $\mathsf{E}_{i,j-1}E_{j-1}$ has zero coefficient at $j-1$. The latter weight statement follows by induction directly from (27). This proves the upper-triangular formula and completes the proof. □

7 The diagonal quotient

Theorem 7.1. *Let I be the two-sided ideal of $\overline{\mathscr{S}}_n(\mathfrak{B}^*)$ generated by*

$$b_{aa}c_{aa} - 1, \qquad 1 \le a \le n.$$

Then

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I \cong \mathcal{U}_q(\mathfrak{sl}_n),$$

where the quantum group is the simply-connected one.

Writing bars for classes in the quotient, the inverse isomorphism from $\mathcal{U}_q(\mathfrak{sl}_n)$ is explicitly

$$\widehat{K}_a \longmapsto \overline{c}_{aa} = \overline{b}_{aa}^{-1}, \tag{36}$$

$$E_i \longmapsto -\lambda^{-1}\overline{c}_{i,i+1}\overline{c}_{i+1,i+1}^{-1}, \tag{37}$$

$$F_i \longmapsto \lambda^{-1}\overline{b}_{i+1,i+1}^{-1}\overline{b}_{i+1,i}. \tag{38}$$

Proof. By Theorem 6.1,

$$\Phi(b_{aa}c_{aa} - 1) = (\widehat{K}_a^-)^{-1}\widehat{K}_a^+ - 1.$$

Since $\widehat{K}_a^-$ is a unit, these elements generate the same ideal of D as

$$\widehat{K}_a^+ - \widehat{K}_a^-, \qquad 1 \le a \le n.$$

Denote that ideal by J. Hence $\Phi(I) = J$ and

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I \cong D/J. \tag{39}$$

There is a surjective homomorphism

$$\pi : D \longrightarrow \mathcal{U}_q(\mathfrak{sl}_n)$$

sending both $\widehat{K}_a^-$ and $\widehat{K}_a^+$ to $\widehat{K}_a$ and sending E_i, F_i to the identically named generators. To check it, use Lemma 5.2: the only new commutator becomes

$$[E_i, F_j] = \delta_{ij}\frac{K_i - K_i^{-1}}{\lambda},$$

and all the other relations become (2)–(6). Clearly $J \subseteq \ker\pi$.

Conversely, in D/J the two toral copies agree. Their common images together with the images of E_i, F_i satisfy the entire presentation (2)–(6). They therefore define a homomorphism

$$\mathcal{U}_q(\mathfrak{sl}_n) \longrightarrow D/J.$$

This homomorphism and the map induced by π are inverse on every generator. Consequently

$$D/J \cong \mathcal{U}_q(\mathfrak{sl}_n) \qquad \text{and} \qquad \ker\pi = J.$$

Together with (39), this proves the claimed quotient, including the assertion that no further relations have been introduced. Formulas (36)–(38) are the specializations of (30)–(31). □

Remark 7.2. The elements $b_{aa}c_{aa}$ are central. Under Φ they become $(\widehat{K}_a^-)^{-1}\widehat{K}_a^+$, and the two toral copies have identical conjugation actions on every E_i and F_i. Furthermore,

$$\prod_{a=1}^{n} b_{aa}c_{aa} = 1.$$

Thus the quotient identifies the two full weight-lattice tori; there are only $n-1$ independent diagonal identifications. This is also why the conclusion is the simply-connected quantum group rather than its root-lattice form.

Remark 7.3 (Parameter qualification). The factors λ^{-1} are essential to the identification with the usual Chevalley generators. At $q = \pm 1$ the skew pairing displayed in the question itself is undefined. Statements at such parameters require an integral-form reformulation and are not obtained by substituting into the formulas above.

References

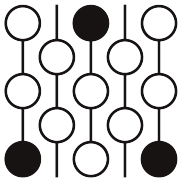
[1] F. Gavarini, *Presentation by Borel subalgebras and Chevalley generators for quantum enveloping algebras*, Proc. Edinburgh Math. Soc. (2) **49** (2006), no. 2, 291–308. See also arXiv:math/0403346v3; the section numbering cited above is that of this version.

[2] A. Klimyk and K. Schmüdgen, *Quantum Groups and Their Representations*, Texts and Monographs in Physics, Springer-Verlag, Berlin, 1997.

[3] T. T. Q. Lê and A. S. Sikora, *Stated* $\mathrm{SL}(n)$*-skein modules and algebras*, J. Topol. **17** (2024), no. 3, e12350. See also arXiv:2201.00045; the theorem and corollary numbering cited above refers to the 85-page arXiv text.

[4] T. T. Q. Lê and T. Yu, *Quantum traces for* SL_n*-skein algebras*, arXiv:2303.08082v2, May 28, 2025.

Review of Problem 3 – Astra



First Proof Third Batch
Reviewer 2
October 8–9, 2026

1 Review

The solution is mostly correct. The strategy is clear and is similar to the human solution and to one of the other submissions. The reduced stated skein algebra of the punctured bigon is identified, as a vector space, with a tensor product of two quotients of the quantum coordinate algebra $\mathcal{O}_q(SL_n)$. These tensor factors are identified with the Borel subalgebras $\mathcal{U}_q(\mathfrak{b}_-)$ and $\mathcal{U}_q(\mathfrak{b}_+)$ of $\mathcal{U}_q(\mathfrak{sl}_n)$. A presentation of the generalized quantum double D is established and used to define an algebra homomorphism Ψ from D to the reduced stated skein algebra of the punctured bigon. That Ψ is an isomorphism follows from identifying it as the composite of three linear isomorphisms.

There are a few places where a statement is correct but the justification is lacking, most notably in the discussions of the boundary-exchange relation and saturated systems of arcs, where it is not entirely clear how the relations in the stated skein algebra and earlier results in the literature are being used. An expert could modify the justifications, but a non-expert could struggle to understand the intent of what is written.

The exposition is clear for the most part, but there are instances where it could be improved. This subject is inherently diagrammatic, and including pictures would greatly aid the exposition (this comment applies to all the submissions). The discussions of the boundary-exchange relation and saturated systems of arcs are unclear; they require more precise references and rephrasing some arguments

At minimum, references to work of Lê, *Triangular decomposition of skein algebras* and Korinman, *Quantum holonomic link invariants derived from stated skein algebras*, should be included. Korinman established the isomorphisms in the problem in the case $n = 2$.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

We give an explicit identification of the reduced stated SL_n-skein algebra of the once-punctured bigon with the generalized quantum double of the two simply-connected quantum Borel algebras. The identification is specified on every horizontal matrix-coefficient arc. Its injectivity follows from an ordered two-arc factorization of the skein algebra, rather than from a comparison of defining relations alone. The prescribed diagonal quotient then identifies the two Cartan tori and gives the simply-connected quantized enveloping algebra.

3 Coefficients and conventions

Fix $n \geq 2$. We give the proof over

$$\mathbb{K} = \mathbb{Q}(v), \qquad q = v^{2n}, \qquad \lambda = q - q^{-1}.$$

Thus all fractional powers needed in the stated skein relations are available, and λ is invertible. The same argument applies at non-root-of-unity specializations over a common ground field containing the required fractional powers. Throughout, $\mathcal{U}_q(\mathfrak{sl}_n)$ means the unrestricted, simply-connected quantum group, not a small quantum group.

We use the stated skein conventions of Lê–Sikora [3]: the product xy places x above y. The reduced algebra is the quotient by the bad corner arcs, as in [4, Section 7.1].

The surface $\mathfrak{B}^*$ is a disk with its top and bottom boundary points and one interior point removed. We fix the following labeling of the horizontal arcs. Both arcs point to the right; the first index is the left state and the second is the right state.

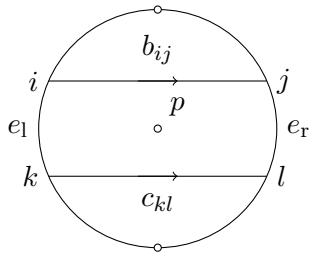


Figure 1: The labeling used in the formulas below.

At the upper corner the bad right-directed arcs are b_{ij} with $i < j$. At the lower corner they are c_{ij} with $i > j$. Consequently,

$$b_{ij} = 0 \quad (i < j), \qquad c_{ij} = 0 \quad (i > j) \qquad \text{in } \overline{\mathscr{S}}_n(\mathfrak{B}^*). \tag{1}$$

The reversal of the inequality at the lower corner is due to the 180° rotation identifying that corner with the standard upper-corner picture. If the two letters in a different drawing are interchanged, the formulas below are to be read with that interchange.

For the quantum group, put

$$\omega_{ai} = \delta_{a,i} - \delta_{a,i+1}, \qquad K_i = \widehat{K}_i \widehat{K}_{i+1}^{-1}.$$

The simply-connected presentation uses

$$\widehat{K}_a^{\pm 1} \quad (1 \le a \le n), \qquad E_i, F_i \quad (1 \le i < n),$$

with commuting toral generators and

$$\prod_{a=1}^{n} \widehat{K}_a = 1, \tag{2}$$

$$\widehat{K}_a E_i \widehat{K}_a^{-1} = q^{\omega_{ai}} E_i, \qquad \widehat{K}_a F_i \widehat{K}_a^{-1} = q^{-\omega_{ai}} F_i, \tag{3}$$

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{\lambda}. \tag{4}$$

For $X = E$ and $X = F$ the remaining relations are

$$X_i X_j = X_j X_i \qquad (|i-j| > 1), \tag{5}$$

$$X_i^2 X_j - (q + q^{-1}) X_i X_j X_i + X_j X_i^2 = 0 \qquad (|i-j| = 1). \tag{6}$$

Our coproduct convention is

$$\begin{aligned} \Delta(\widehat{K}_a) &= \widehat{K}_a \otimes \widehat{K}_a, \\ \Delta(E_i) &= E_i \otimes K_i + 1 \otimes E_i, \\ \Delta(F_i) &= F_i \otimes 1 + K_i^{-1} \otimes F_i. \end{aligned} \tag{7}$$

These conventions specify the extended quantum group in the question; see [2]. The corresponding Borel presentations use the toral relations and the respective E- or F-relations. They also follow from the quantum-matrix realizations recalled in Lemma 5.1 below.

4 The skein-theoretic matrix calculation

Let $V = \mathbb{K}^n$, let e_{ab} denote its matrix units, and set

$$R = q^{-1/n} \left(\sum_{a,b=1}^{n} q^{\delta_{ab}} e_{aa} \otimes e_{bb} + \lambda \sum_{a>b} e_{ab} \otimes e_{ba} \right). \tag{8}$$

The braiding matrix is PR, where P interchanges the tensor factors. This is the normalization used in [3, Equation (9)].

For a matrix $T = (t_{ij})$, write $T_1 = T \otimes I_n$ and $T_2 = I_n \otimes T$. The standard quantum coordinate algebra $\mathcal{O}_q(\mathrm{SL}_n)$ has presentation

$$R T_1 T_2 = T_2 T_1 R, \qquad \det_q(T) = 1, \tag{9}$$

where

$$\det_q(T) = \sum_{\sigma \in S_n} (-q)^{\ell(\sigma)} t_{1,\sigma(1)} \cdots t_{n,\sigma(n)}.$$

> **Reviewer 2**
>
> This definition gives relations but does not explicitly say what the generators are.

Define its triangular quotients

$$\mathcal{T}^- = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i < j), \qquad \mathcal{T}^+ = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i > j). \tag{10}$$

Lemma 4.1 (Two-arc matrix calculation)**.** *Let $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$, and let $B = (b_{ij})$ and $C = (c_{ij})$ be as in Figure 1.*

1. *B and C satisfy* (9)*. Moreover, B is lower triangular and C is upper triangular.*
2. *The mixed relation is*
$$RB_1C_2 = C_2B_1R. \tag{11}$$
Equivalently, for all i, j, k, l,
$$q^{\delta_{ik}} b_{ij}c_{kl} - q^{\delta_{jl}} c_{kl}b_{ij} = \lambda\big(\mathbf{1}_{j<l} c_{kj}b_{il} - \mathbf{1}_{k<i} b_{kj}c_{il}\big). \tag{12}$$
3. *The homomorphisms from $\mathcal{T}^-$ and $\mathcal{T}^+$ determined by $T \mapsto B$ and $T \mapsto C$ are injective, and multiplication induces a vector-space isomorphism*
$$\mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ \xrightarrow{\sim} A, \qquad f \otimes g \longmapsto f(B)g(C). \tag{13}$$
This is a vector-space assertion, not an assertion that the ordinary tensor-product multiplication is preserved.

Proof. A thin neighborhood of either horizontal arc is an ideal bigon. The bigon theorem [3, Theorem 6.3] identifies its stated skein algebra with $\mathcal{O}_q(\mathrm{SL}_n)$, with the right-directed stated arcs as the matrix generators. This proves the quantum-matrix and determinant relations. Equation (1) gives the asserted triangularity.

Here are the boundary-exchange details for the mixed relation. They hold already in the unreduced algebra. Consider the two disjoint arcs with c higher than b at the left boundary and b higher than c at the right boundary. Denote the resulting mixed-height diagram, with external state pairs (i, k) and (j, l), by $W_{ik;jl}$. Applying the boundary height-exchange relation at the left or at the right gives, respectively,

$$W_{ik;jl} = \sum_{a,b} R_{ik,ab}\, b_{aj}c_{bl} = \sum_{a,b} c_{kb}b_{ia}\, R_{ab,jl}. \tag{14}$$

Here the flip in the braiding PR has been removed by recording the indices in the order of the two arcs. Thus the coefficients appearing in the two exchanges are

$$R_{ik,ab} = q^{-1/n}\left(q^{\delta_{ik}}\delta_{ia}\delta_{kb} + \lambda \mathbf{1}_{k<i}\delta_{ka}\delta_{ib}\right),$$
$$R_{ab,jl} = q^{-1/n}\left(q^{\delta_{jl}}\delta_{aj}\delta_{bl} + \lambda \mathbf{1}_{j<l}\delta_{al}\delta_{bj}\right).$$

These are precisely the boundary braiding coefficients (8); the boundary relations and their braiding interpretation are those of [3, Section 4 and Theorem 6.4]. Substitution into (14), followed by cancellation of $q^{-1/n}$, proves (12).

> **Reviewer 2**
>
> This discussion of the boundary-exchange relations would be greatly helped by a picture, a more precise reference than [3, Section 4], and an explanation of how the braiding (8) fits into the relations in the stated skein algebra.

It remains to prove the vector-space assertion, including the absence of additional relations after reduction. Write $S = \mathscr{S}_n(\mathfrak{B}^*)$ for the unreduced algebra, and temporarily use b, c for the unreduced arcs. They form the standard saturated two-arc system for $\mathfrak{B}^*$: together with the boundary edges they give a groupoid spine of the punctured bigon. The ordered-arc factorization theorem, with the b-neighborhood above the c-neighborhood at both boundary edges, gives

$$m : \mathcal{O}_q(\mathrm{SL}_n) \otimes_{\mathbb{K}} \mathcal{O}_q(\mathrm{SL}_n) \xrightarrow{\sim} S, \qquad f \otimes g \longmapsto f(B)g(C). \tag{15}$$

The relevant result permits arbitrary orders at the boundary edges; see [3, Corollary 8.14].

> **Reviewer 2**
>
> It is not clear what is meant by the two sentences above Equation (15). This needs a more clear explanation of the relevant parts of [3, Section 8.4], which discusses saturated systems of arcs.

Let J_b be the ideal of the first copy of $\mathcal{O}_q(\mathrm{SL}_n)$ generated by t_{ij} with $i < j$, and let J_c be the ideal of the second copy generated by t_{ij} with $i > j$. Let I_{top} and I_{bottom} be the respective bad-corner ideals in S.

The bad-arc normality theorem [4, Theorem 7.2, especially Equation (128)] says that each corner ideal is both the left and the right ideal generated by either one of its two oriented families of bad arcs. In our notation this gives

$$I_{\mathrm{top}} = \mathrm{span}_{\mathbb{K}}\{b_{ij} : i < j\}\, S, \qquad I_{\mathrm{bottom}} = S\, \mathrm{span}_{\mathbb{K}}\{c_{ij} : i > j\}.$$

In particular, reversed orientations introduce no further ideal.

Using (15), the first equality implies

$$I_{\mathrm{top}} \subseteq m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The reverse inclusion holds because I_{top} is a two-sided ideal containing the generators of J_b. Hence

$$I_{\mathrm{top}} = m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The same argument, using the displayed right-ideal description at the lower corner, gives

$$I_{\mathrm{bottom}} = m(\mathcal{O}_q(\mathrm{SL}_n) \otimes J_c).$$

Therefore

$$I^{\mathrm{bad}} = m\big(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n) + \mathcal{O}_q(\mathrm{SL}_n) \otimes J_c\big).$$

Taking the quotient of the vector-space isomorphism (15) proves (13). The injectivity of the two factor maps follows as well. □

We record two elementary consequences that will also make the eventual isomorphism explicit on all the arcs. The diagonal entries of either triangular quantum matrix commute, and

$$\prod_{a=1}^{n} b_{aa} = \prod_{a=1}^{n} c_{aa} = 1. \tag{16}$$

Thus every diagonal entry is invertible. Expanding the quantum-matrix relation on the indicated 2×2 submatrices gives, for $i < j-1$,

$$b_{ji} = \lambda^{-1} b_{j-1,j-1}^{-1} [b_{j-1,i}, b_{j,j-1}], \tag{17}$$

$$c_{ij} = \lambda^{-1} c_{j-1,j-1}^{-1} [c_{i,j-1}, c_{j-1,j}]. \tag{18}$$

Indeed, these are the relations $ad - da = \lambda bc$ for rows $j-1, j$ and columns $i, j-1$, and for rows $i, j-1$ and columns $j-1, j$, respectively. In each case the diagonal factor on the right commutes with the opposite off-diagonal factor. Consequently, each triangular algebra is generated by its diagonal and immediately adjacent off-diagonal entries.

5 The quantum Borel algebras and their double

5.1 Triangular matrix realizations

Write

$$H_- = \mathcal{U}_q(\mathfrak{b}_-), \qquad H_+ = \mathcal{U}_q(\mathfrak{b}_+).$$

Lemma 5.1 (Borel matrix realizations)**.** *There are algebra isomorphisms*

$$\eta_- : H_- \xrightarrow{\sim} \mathcal{T}^-, \qquad \eta_+ : H_+ \xrightarrow{\sim} \mathcal{T}^+$$

given by

$$\eta_-(\widehat{K}_a) = b_{aa}^{-1}, \qquad \eta_-(F_i) = \lambda^{-1} b_{i+1,i+1}^{-1} b_{i+1,i}, \tag{19}$$

$$\eta_+(\widehat{K}_a) = c_{aa}, \qquad \eta_+(E_i) = -\lambda^{-1} c_{i,i+1} c_{i+1,i+1}^{-1}. \tag{20}$$

Proof. For $\mathfrak{gl}_n$, with independent toral generators $G_1, \dots, G_n$, these are exactly the algebra isomorphisms ζ_- and ζ_+ in [1, Section 2.4]. Their inverses on adjacent entries are

$$t_{i+1,i} \longmapsto \lambda G_{i+1}^{-1} F_i, \qquad t_{i,i+1} \longmapsto -\lambda E_i G_{i+1},$$

and the diagonal entries map to G_a^{-1} and G_a, respectively.

Impose the central relation $\prod_a G_a = 1$. On the matrix sides its image is the relation that the product of the diagonal entries is 1. For a triangular matrix this is exactly the quantum determinant-one relation. Taking these quotients gives (19)–(20). The toral lattice is $\mathbb{Z}^n/\mathbb{Z}(1, \dots, 1)$, namely the weight lattice of type A_{n-1}, so the resulting Borel algebras are the simply-connected ones.

Only the algebra assertion of [1, Section 2.4] is used: the coproduct convention there is opposite to (7). □

5.2 A presentation of the generalized double

Let

$$D = H_- \bowtie H_+$$

be the generalized quantum double for the pairing in the question. We denote its two toral copies by $\widehat{K}_a^-$ and $\widehat{K}_a^+$, and set

$$K_i^- = \widehat{K}_i^- (\widehat{K}_{i+1}^-)^{-1}, \qquad K_i^+ = \widehat{K}_i^+ (\widehat{K}_{i+1}^+)^{-1}.$$

The superscript $-$ labels the negative Borel copy; it does not mean an inverse.

For clarity, we specify the double convention. The skew pairing $\tau : H_+ \otimes H_- \to \mathbb{K}$ satisfies

$$\tau(xx', y) = \tau(x, y_{(1)})\tau(x', y_{(2)}),$$
$$\tau(x, yy') = \tau(x_{(1)}, y')\tau(x_{(2)}, y),$$

and has generator values

$$\begin{aligned} \tau(\widehat{K}_a, \widehat{K}_b) &= q^{1/n-\delta_{ab}}, \\ \tau(E_i, F_j) &= -\delta_{ij}\lambda^{-1}, \\ \tau(\widehat{K}_a, F_j) &= \tau(E_i, \widehat{K}_b) = 0. \end{aligned} \tag{21}$$

The underlying vector space of D is $H_- \otimes H_+$, with multiplication identifying each factor with a subalgebra. Its exchange convention is

$$\sum \tau(x_{(1)}, y_{(1)})x_{(2)}y_{(2)} = \sum y_{(1)}x_{(1)}\tau(x_{(2)}, y_{(2)}), \qquad x \in H_+, \quad y \in H_-. \tag{22}$$

Writing this convention explicitly avoids ambiguity with opposite-double conventions.

Lemma 5.2 (Presentation of the double)**.** *The algebra D is presented by the two Borel presentations and the following mixed relations:*

$$[\widehat{K}_a^+, \widehat{K}_b^-] = 0, \tag{23}$$

$$\widehat{K}_a^+ F_i (\widehat{K}_a^+)^{-1} = q^{-\omega_{ai}} F_i, \qquad \widehat{K}_a^- E_i (\widehat{K}_a^-)^{-1} = q^{\omega_{ai}} E_i, \tag{24}$$

$$[E_i, F_j] = \delta_{ij} \frac{K_i^+ - (K_i^-)^{-1}}{\lambda}. \tag{25}$$

Proof. First derive the relations in D. For the toral terms,

$$\tau(\widehat{K}_a, K_i) = \frac{q^{1/n-\delta_{ai}}}{q^{1/n-\delta_{a,i+1}}} = q^{-\omega_{ai}}.$$

Substitution into (22) proves (23) and (24).

For $x = E_i$ and $y = F_j$, the left and right sides of (22), using (7), are

$$E_iF_j - \delta_{ij}\lambda^{-1}K_i^+, \qquad F_jE_i - \delta_{ij}\lambda^{-1}(K_j^-)^{-1},$$

respectively. This proves (25).

To see that these are a complete set of mixed relations, let D_0 be the algebra given by the two Borel presentations and (23)–(25). The derived relations give a surjection $D_0 \to D$.

They also allow every word in D_0 to be written as a sum of words with the negative Borel generators first and the positive Borel generators second. Indeed, the toral exchanges merely introduce scalars; the exchange of E_i with F_j has a reordered term and terms with fewer E's and F's. Induction first on the number of such generators and then on the number of mixed inversions proves the assertion. Hence multiplication gives a surjection

$$H_- \otimes H_+ \longrightarrow D_0.$$

Its composition with $D_0 \to D$ is the defining vector-space isomorphism $H_- \otimes H_+ \to D$. It follows that both maps are isomorphisms, proving the presentation. □

6 The explicit isomorphism

Define elements of the two Borel factors, for $1 \le i < j \le n$, by

$$\mathsf{F}_{i,i+1} = F_i, \qquad \mathsf{F}_{ij} = \mathsf{F}_{i,j-1}F_{j-1} - q^{-1}F_{j-1}\mathsf{F}_{i,j-1} \qquad (j > i+1), \tag{26}$$

$$\mathsf{E}_{i,i+1} = E_i, \qquad \mathsf{E}_{ij} = E_{j-1}\mathsf{E}_{i,j-1} - q\,\mathsf{E}_{i,j-1}E_{j-1} \qquad (j > i+1). \tag{27}$$

These recursions fix all root-vector normalizations needed below.

Theorem 6.1. *There is an algebra isomorphism*

$$\Phi : \overline{\mathscr{S}}_n(\mathfrak{B}^*) \xrightarrow{\sim} \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$$

given on the horizontal arcs by

$$\Phi(b_{aa}) = (\widehat{K}_a^-)^{-1}, \qquad \Phi(c_{aa}) = \widehat{K}_a^+, \tag{28}$$

$$\Phi(b_{ji}) = \lambda(\widehat{K}_j^-)^{-1}\mathsf{F}_{ij}, \qquad \Phi(c_{ij}) = -\lambda\mathsf{E}_{ij}\widehat{K}_j^+ \qquad (i < j). \tag{29}$$

The other off-diagonal arcs vanish by (1).

Equivalently, the inverse isomorphism $\Psi = \Phi^{-1}$ is determined by

$$\Psi(\widehat{K}_a^-) = b_{aa}^{-1}, \qquad \Psi(\widehat{K}_a^+) = c_{aa}, \tag{30}$$

$$\Psi(F_i) = \lambda^{-1}b_{i+1,i+1}^{-1}b_{i+1,i}, \qquad \Psi(E_i) = -\lambda^{-1}c_{i,i+1}c_{i+1,i+1}^{-1}. \tag{31}$$

Proof. We first construct Ψ and prove that it is an isomorphism.

For brevity put, in $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$,

$$d_a = b_{aa}, \qquad t_a = c_{aa}, \qquad x_i = b_{i+1,i}, \qquad y_i = c_{i,i+1}.$$

The diagonal elements are invertible by (16). Lemma 5.1 gives algebra homomorphisms from the two Borel factors to A with the values (30)–(31). Thus it remains to check the mixed relations in Lemma 5.2.

The internal quantum-matrix relations and (12) give

$$[d_a, d_b] = [t_a, t_b] = [d_a, t_b] = 0 \tag{32}$$

and

$$\begin{aligned} d_a x_i &= q^{\omega_{ai}} x_i d_a, \quad & t_a x_i &= q^{-\omega_{ai}} x_i t_a, \\ d_a y_i &= q^{-\omega_{ai}} y_i d_a, \quad & t_a y_i &= q^{\omega_{ai}} y_i t_a. \end{aligned} \tag{33}$$

For example, setting the first matrix entry in (12) equal to b_{aa} makes both correction terms vanish by lower triangularity, and yields

$$d_a c_{kl} = q^{\delta_{al}-\delta_{ak}} c_{kl} d_a.$$

Setting the second entry equal to c_{aa} gives

$$t_a b_{kl} = q^{\delta_{ak}-\delta_{al}} b_{kl} t_a.$$

These are the mixed assertions in (33); the internal ones are obtained in the same way from the single-matrix relation.

Write

$$\mathcal{F}_i = \lambda^{-1} d_{i+1}^{-1} x_i, \qquad \mathcal{E}_i = -\lambda^{-1} y_i t_{i+1}^{-1}.$$

Equations (32) and (33) immediately give

$$t_a \mathcal{F}_i t_a^{-1} = q^{-\omega_{ai}} \mathcal{F}_i, \qquad d_a^{-1} \mathcal{E}_i d_a = q^{\omega_{ai}} \mathcal{E}_i.$$

Thus the mixed toral relations are satisfied.

For the remaining mixed relation, substitute

$$(i,k;j,l) = (j+1, i; j, i+1)$$

into (12). Triangularity makes both correction terms vanish unless $i = j$, so that

$$q^{\delta_{i,j+1}} x_j y_i - q^{\delta_{j,i+1}} y_i x_j = \lambda \delta_{ij} (d_{i+1} t_i - d_i t_{i+1}). \tag{34}$$

Using (33) to move the diagonal factors gives

$$\begin{aligned} [\mathcal{E}_i, \mathcal{F}_j] &= \lambda^{-2} d_{j+1}^{-1} \left(x_j y_i - q^{\delta_{j,i+1} - \delta_{i,j+1}} y_i x_j \right) t_{i+1}^{-1} \\ &= \delta_{ij} \lambda^{-1} d_{i+1}^{-1} (d_{i+1} t_i - d_i t_{i+1}) t_{i+1}^{-1} \\ &= \delta_{ij} \frac{t_i t_{i+1}^{-1} - d_i d_{i+1}^{-1}}{\lambda}. \end{aligned} \tag{35}$$

But under the proposed map,

$$K_i^+ \longmapsto t_i t_{i+1}^{-1}, \qquad (K_i^-)^{-1} \longmapsto d_i d_{i+1}^{-1}.$$

Therefore (35) is precisely (25). Lemma 5.2 now supplies an algebra homomorphism

$$\Psi : D \longrightarrow A$$

with the required values.

We prove its bijectivity without making an unsupported presentation claim about A. The following diagram commutes:

$$\begin{array}{ccc} H_- \otimes_{\mathbb{K}} H_+ & \xrightarrow{m_D} & D \\ {\scriptstyle \eta_- \otimes \eta_+} \big\downarrow \cong & & \big\downarrow {\scriptstyle \Psi} \\ \mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ & \xrightarrow{m_A} & A. \end{array}$$

The top arrow is the defining vector-space isomorphism for the generalized double. The left arrow is an isomorphism by Lemma 5.1, and the bottom arrow is an isomorphism by Lemma 4.1. Consequently Ψ is a vector-space isomorphism and hence an algebra isomorphism.

It remains to verify the formulas for its inverse on the non-adjacent entries. The inverse formulas on the diagonal and adjacent entries follow directly from (30)–(31). For completeness, the induction using (17) and (18) is as follows.

Put

$$m_a = (\widehat{K}_a^-)^{-1}, \qquad p_a = \widehat{K}_a^+.$$

For $i < j-1$, assume the asserted formulas at distance $j-i-1$. The toral weight relations give

$$\lambda^{-1}m_{j-1}^{-1}[\,\lambda m_{j-1}\mathsf{F}_{i,j-1},\lambda m_jF_{j-1}\,]$$
$$=\lambda m_j\big(\mathsf{F}_{i,j-1}F_{j-1}-q^{-1}F_{j-1}\mathsf{F}_{i,j-1}\big)=\lambda m_j\mathsf{F}_{ij}.$$

Here m_j commutes with $\mathsf{F}_{i,j-1}$ and $F_{j-1}m_{j-1}=q^{-1}m_{j-1}F_{j-1}$. This proves the lower-triangular formula.

Similarly,

$$\lambda^{-1}p_{j-1}^{-1}[\,-\lambda\mathsf{E}_{i,j-1}p_{j-1},-\lambda E_{j-1}p_j\,]$$
$$=-\lambda\big(E_{j-1}\mathsf{E}_{i,j-1}-q\mathsf{E}_{i,j-1}E_{j-1}\big)p_j=-\lambda\mathsf{E}_{ij}p_j.$$

Indeed, p_j commutes with $\mathsf{E}_{i,j-1}$, $p_{j-1}E_{j-1}=qE_{j-1}p_{j-1}$, and the total weight of $\mathsf{E}_{i,j-1}E_{j-1}$ has zero coefficient at $j-1$. The latter weight statement follows by induction directly from (27). This proves the upper-triangular formula and completes the proof. □

7 The diagonal quotient

Theorem 7.1. *Let I be the two-sided ideal of $\overline{\mathscr{S}}_n(\mathfrak{B}^*)$ generated by*

$$b_{aa}c_{aa}-1,\qquad 1\le a\le n.$$

Then

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I\cong\mathcal{U}_q(\mathfrak{sl}_n),$$

where the quantum group is the simply-connected one.

Writing bars for classes in the quotient, the inverse isomorphism from $\mathcal{U}_q(\mathfrak{sl}_n)$ is explicitly

$$\widehat{K}_a\longmapsto\overline{c}_{aa}=\overline{b}_{aa}^{-1},\tag{36}$$
$$E_i\longmapsto-\lambda^{-1}\overline{c}_{i,i+1}\overline{c}_{i+1,i+1}^{-1},\tag{37}$$
$$F_i\longmapsto\lambda^{-1}\overline{b}_{i+1,i+1}^{-1}\overline{b}_{i+1,i}.\tag{38}$$

Proof. By Theorem 6.1,

$$\Phi(b_{aa}c_{aa}-1)=(\widehat{K}_a^-)^{-1}\widehat{K}_a^+-1.$$

Since $\widehat{K}_a^-$ is a unit, these elements generate the same ideal of D as

$$\widehat{K}_a^+-\widehat{K}_a^-,\qquad 1\le a\le n.$$

Denote that ideal by J. Hence $\Phi(I)=J$ and

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I\cong D/J.\tag{39}$$

There is a surjective homomorphism

$$\pi:D\longrightarrow\mathcal{U}_q(\mathfrak{sl}_n)$$

sending both $\widehat{K}_a^-$ and $\widehat{K}_a^+$ to $\widehat{K}_a$ and sending E_i, F_i to the identically named generators. To check it, use Lemma 5.2: the only new commutator becomes

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{\lambda},$$

and all the other relations become (2)–(6). Clearly $J \subseteq \ker \pi$.

Conversely, in D/J the two toral copies agree. Their common images together with the images of E_i, F_i satisfy the entire presentation (2)–(6). They therefore define a homomorphism

$$\mathcal{U}_q(\mathfrak{sl}_n) \longrightarrow D/J.$$

This homomorphism and the map induced by π are inverse on every generator. Consequently

$$D/J \cong \mathcal{U}_q(\mathfrak{sl}_n) \qquad \text{and} \qquad \ker \pi = J.$$

Together with (39), this proves the claimed quotient, including the assertion that no further relations have been introduced. Formulas (36)–(38) are the specializations of (30)–(31). □

Remark 7.2. The elements $b_{aa}c_{aa}$ are central. Under Φ they become $(\widehat{K}_a^-)^{-1}\widehat{K}_a^+$, and the two toral copies have identical conjugation actions on every E_i and F_i. Furthermore,

$$\prod_{a=1}^{n} b_{aa}c_{aa} = 1.$$

Thus the quotient identifies the two full weight-lattice tori; there are only $n-1$ independent diagonal identifications. This is also why the conclusion is the simply-connected quantum group rather than its root-lattice form.

Remark 7.3 (Parameter qualification). The factors λ^{-1} are essential to the identification with the usual Chevalley generators. At $q = \pm 1$ the skew pairing displayed in the question itself is undefined. Statements at such parameters require an integral-form reformulation and are not obtained by substituting into the formulas above.

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